



传感赋能：半导体创新助力机器人智能进化



像人类一样感知世界

面向消费与汽车应用的创新产品组合

3D Time-of-Flight

Radar 60GHz

Capacitive Sensors

Silicon microphones

Gas sensors



Child presence detection



Passenger classification



Hands On Detection



Driver monitoring



Health monitoring



Intuitive HMI



Noise cancellation



Air quality

应用示例： 商业机器人中的XENSIV™磁性传感器

Current Sensing

- **Motor torque control:**
Detection of force on head, legs, arms, fingers and grippers
- **Charger**
- **BMS**
- **Circuit Breaker**

TLx4971/72/73 – High precision Hall sensor

TLI5570/72 – Cost-effective TMR sensor

Position Sensing

- **Control Elements** (e.g. jog-wheels, emergency stop buttons)
- **Control Joystick**
- **Charging port contact**
- **Gimbal** (camera stabilization)
- **Collision detection**
- **Door/flap/lid open/close contact**

TLx4964x – Hall switch

TLx4966x – Hall switch

TLI5012B – Digital angle sensor

TLx493D-x2B6 – 3D Hall sensor

TLI55910 – Linear TMR sensor

BLDC Motor Commutation

- **Rotor Position Detection in:**
 - Wheel Motor
 - Head, legs, arms, fingers, grippers

TLx4961x – Hall switch

TLE5501 – Angle Sensor "Dual Bridge"

TLI5012B – Digital angle sensor

TLI49012 – Angle sensor EoS/OoS¹

¹ supporting **End of Shaft**, **Out of Shaft** design flexibility

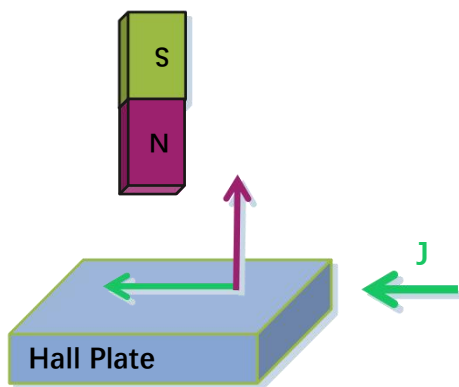
英飞凌 —— 全面提供磁性传感技术，并实现生产

Hall

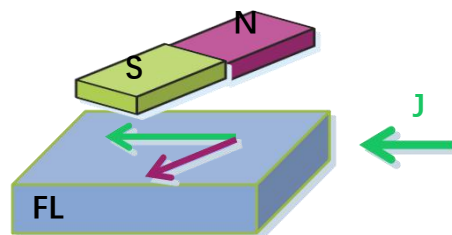
AMR

GMR

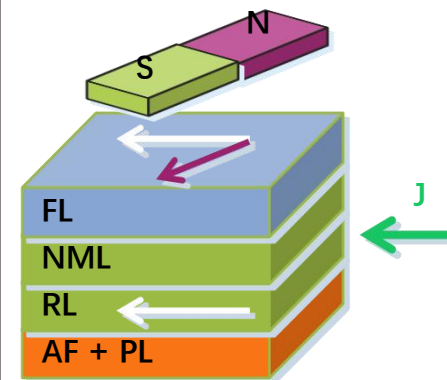
TMR

 ΔV


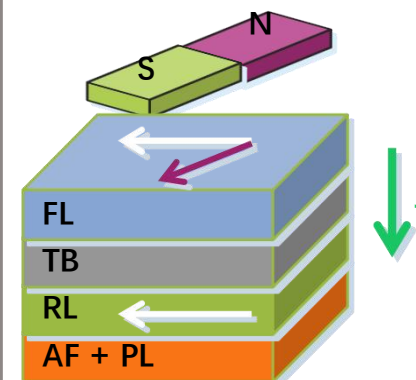
- › High field range
- › No magnetic hysteresis
- › Monolithically integrated in BiCMOS, BCD

 ΔR


- › High sensitivity
- › High field range
- › Low Jitter
- › 180° angle range
- › Monolithically integrated in CMOS, BCD

 ΔR


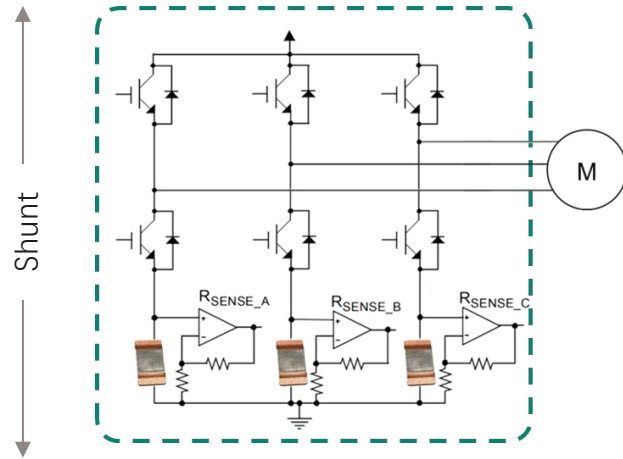
- › Very high sensitivity
- › For fields up to 100mT
- › Low Jitter
- › 360° angle range
- › Monolithically integrated in CMOS, BiCMOS, BCD

 ΔR


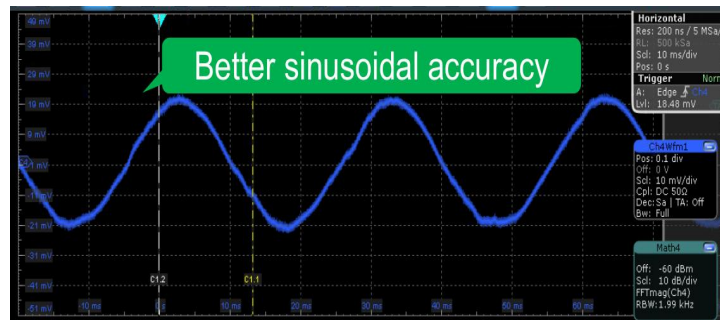
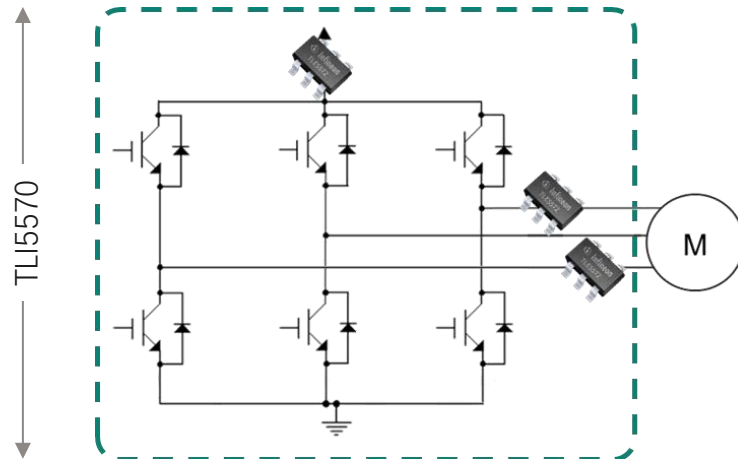
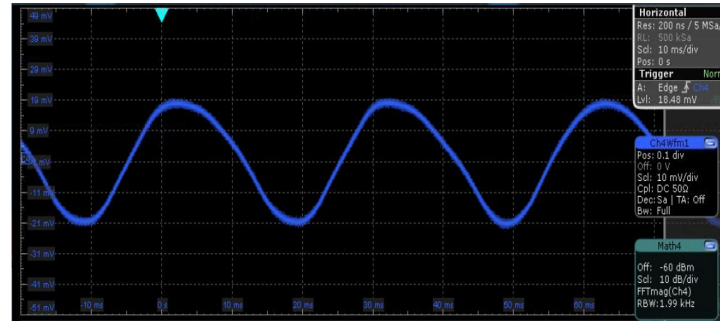
- › Ultra high sensitivity
- › For fields up to 100mT
- › Low Jitter
- › 360° angle range
- › Low power

TLx5570/72在电机控制方面优于Shunt，因为它能以更高的精度和更低的谐波测量相电流

Current measurement topology

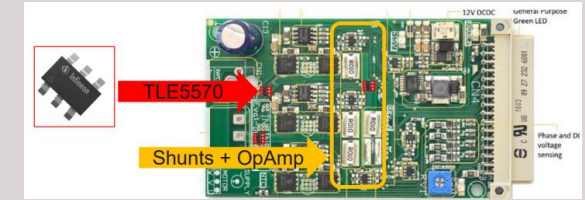


Reconstructed phase current



Test parameters

- ATV SP tested TLI5570 for Motor control using Nanotec DB42S03 4000rpm, 5Amps motor in Motix MiniTest bench

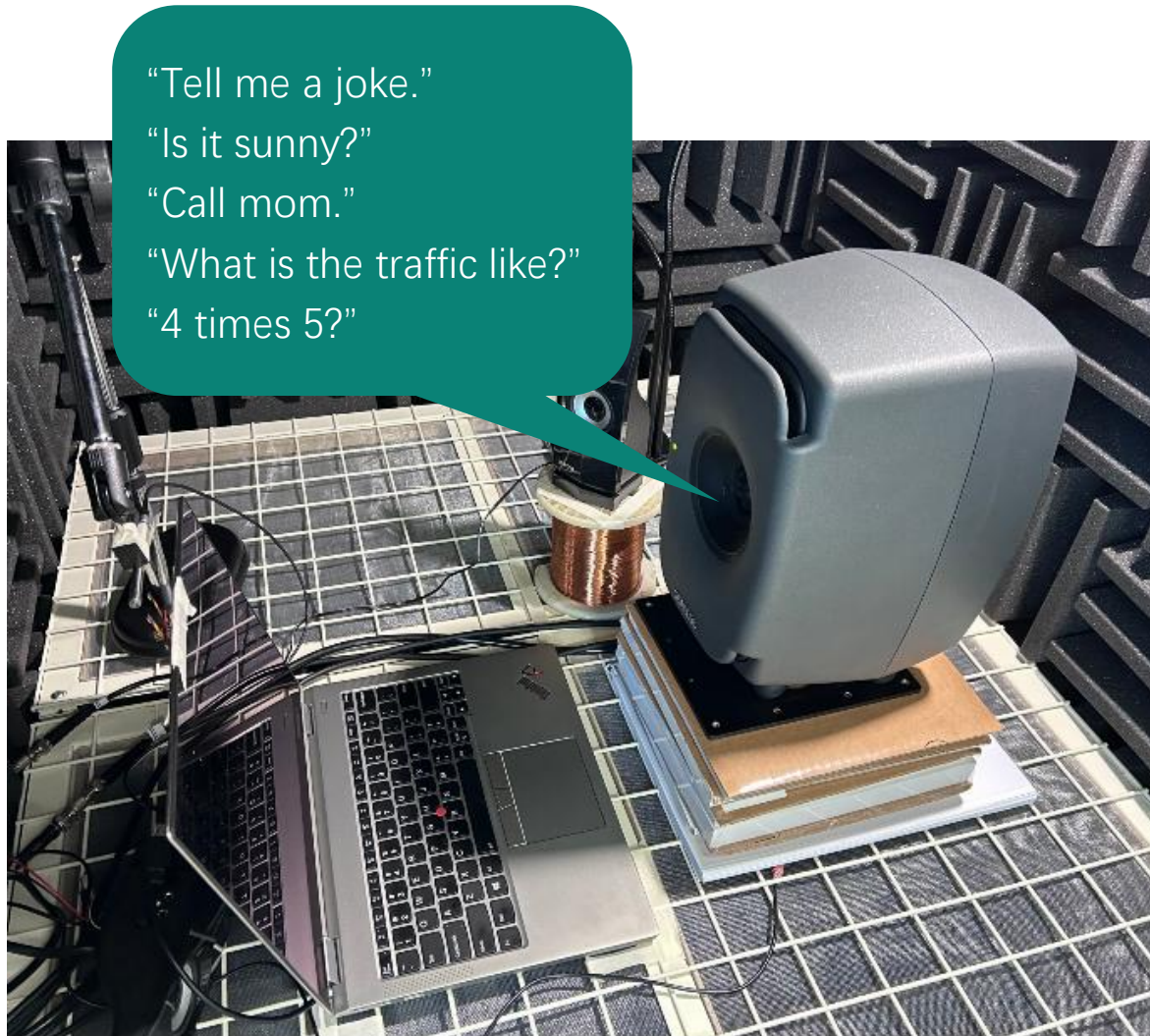


Test results

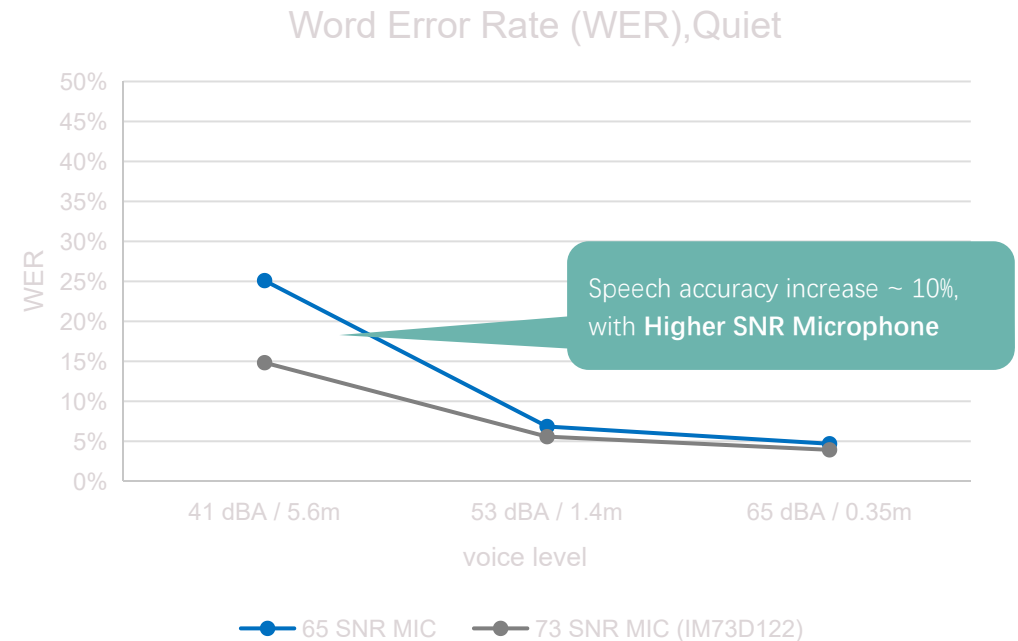
- TLI5570 presented several advantages such as no current sense blind spots, full voltage output, less harmonics
- TLI5570 displayed excellent performance even at low currents, enabling accurate current control and FOC operation

“Using TLE5570(2) to replace Shunts will gain major benefits in multiple application
TLE5570(2) can be used in motor control very well ! - ATV SP test report

OpenAI – Whisper 命令识别在使用更好麦克风时表现更优，尤其在低声压水平下



Voice
Commands



XENSIV™ 麦克风的高性能为全新的音频应用场景提供了可能，并支持多种应用

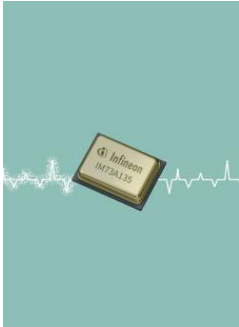


Microphone characteristics



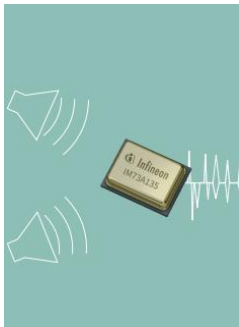
Robust design

Ingress protection (IP57) at microphone level ensures protection against water & dust



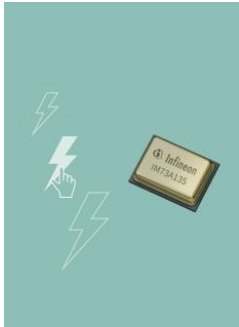
low-noise

Enables best-in-class transparent hearing and active noise cancellation



Wide dynamic range

Stable handling of lowest sounds & highest signal peaks for clear audio capturing



Different power modes

Ultra-low power selectable power modes for long-lasting battery life



Highest quality

Highest qualification standards and 100% test coverage during manufacturing

Advanced audio use-cases

Automated Speech Recognition



Connecting the real to the digital world (for smart speakers, conference systems)

Active Noise Cancellation



Suppression of unwanted sounds (for advanced earbuds, headsets)

Beamforming and Audio Zoom



Changing audio recording directivity (for smartphones, cameras, conference devices)

Transparent / Advanced Hearing



Adaptive audio experience (for hearables and headphones)

高灵敏度气压传感器使机器人更智能



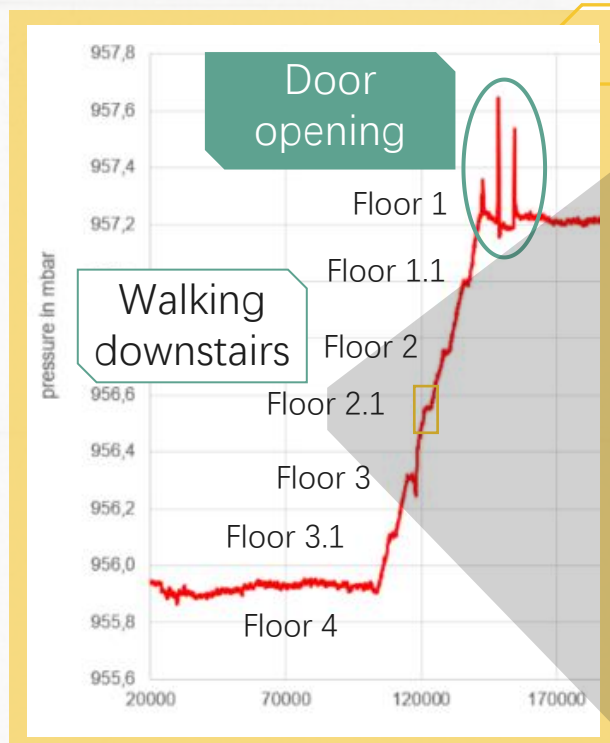
Sensors have a resolution of up to 0.002 hP (2 cm)



Low pressure noise of only ~0.4 Pa

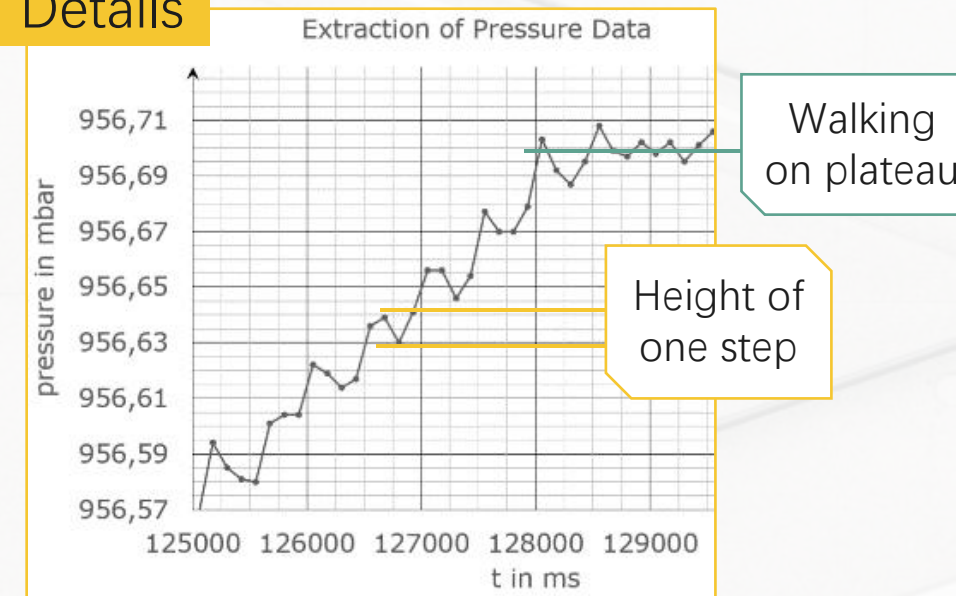


Check which floor the robot is at, without wifi nor vision



A walk through the office with **DPS368**

Details



英飞凌的气压传感器是高精度与低功耗的完美选择



Small size



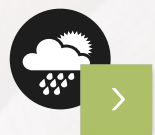
Low energy consumption



Excellent temperature stability



High sensitivity



Robustness



High precision and low power consumption

Waterproof: Fully functional 50 m under water for 1 hour (IPX8)
Protection against water, dust and humidity

用于割草机的雷达表面检测

60 Ghz Radar and robot setup



Technical and Application details

Parameter	Value	Learn more	
Detectable Materials	Grass, No Grass, Cliff, Obstacle	IFX components	BGT60TR13C CY8C6247FDI-D02T (PSOC™ 6 MCU: 150MHz, 1024KB Flash, 288KB RAM) Imagimob Software (Ready Model)
		Focus application	Domestic robots (lawn mower)
Accuracy	>90%		
Performance	+10 Hz embedded on PSOC6		
Model Size	PSOC6 with 1024KB Flash, 288KB RAM		

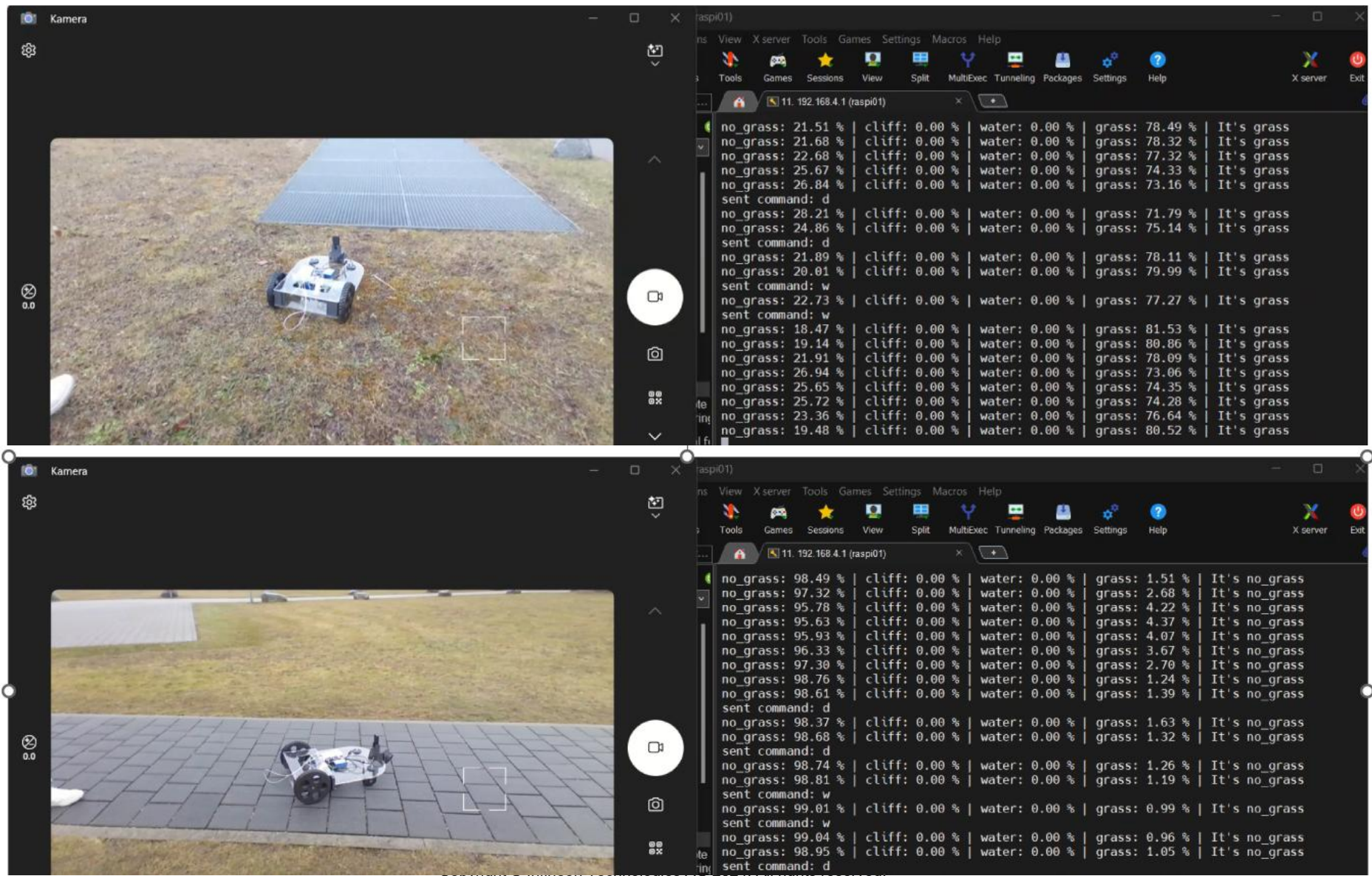
Features

- › Surface, cliff and close-range obstacle detection based on 60 Ghz radar
- › Data collection over several regions to train machine learning model

Benefits

- › High Performance cost-optimized solution (vs. competitive vision-based solutions)
- › Small machine learning model size fits on PSOC6
- › Infineon Hardware and Software offering

演示视频（在 Campeon 测试）



24GHz 与 60GHz 雷达产品组合

60GHz

24GHz

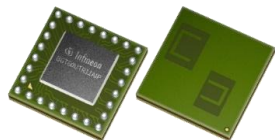
1Tx / 1Rx

PIR replacement



3.3x6.5 mm²

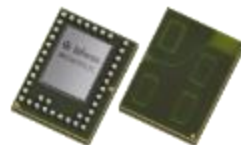
BGT60LTR11AIP
Doppler Radar
Antenna-in-Package



4.05x4.05 mm²

BGT60UTR11AIP
FMCW Radar
Antenna-in-Package

1Tx / 3Rx



5x6.5 mm²

BGT60TR13C
FMCW Radar
Antenna-in-Package

2Tx / 2Rx



3.6 x 3.6 mm

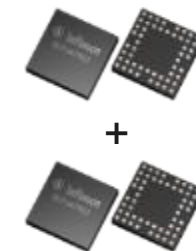
BGT24LTR22

2Tx / 4Rx



6x6 mm²

BGT60ATR24C
FMCW Radar
Without Antenna



Cascading
BGT24LTR22

Time of Flight in Robotic Applications

ToF 传感器在机器人上的应用



Lawn Mowers

- Obstacle avoidance in an open populated environment
- Able to operate in strong sunlight
- Consumer focus = low cost



Robot Vacuum Cleaners

- Long range room mapping (**SLAM**)
- Obstacle avoidance in an open environment
- Need to detect smallest objects
- Exposed to shiny floor
- Consumer focus = low cost



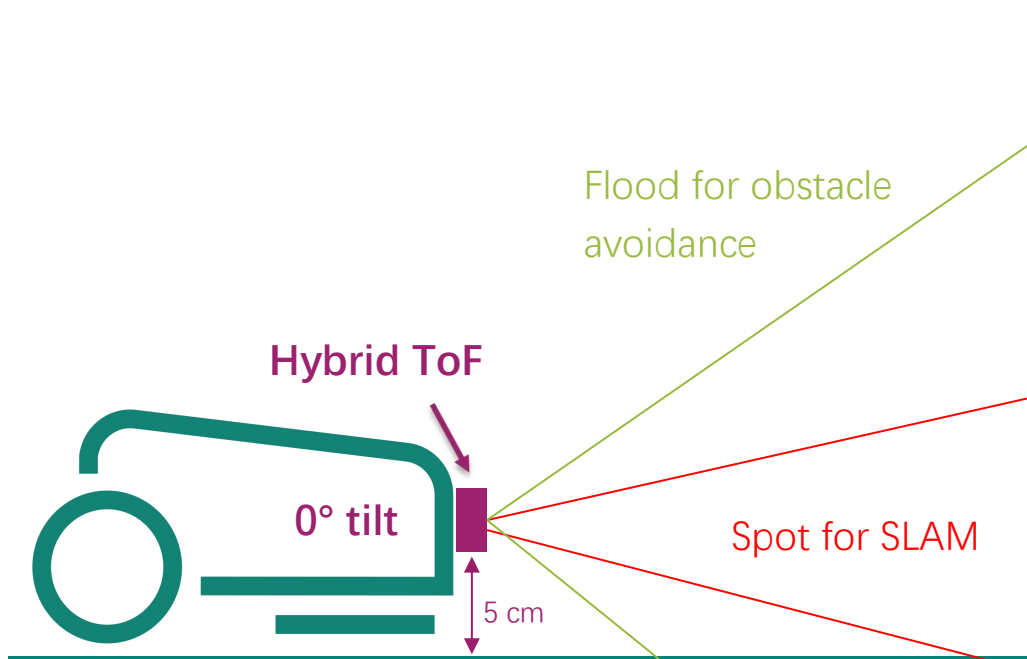
Industrial & Service Robots

- Obstacle avoidance in wide open industrial environment
- Need to detect smallest objects
- High resolution cliff detection
- Heavy machines = **Safety**
- Meet industrial standards

IRS2875C – Hybrid Time of Flight camera 基于IRS2875C的hToF摄像头模组

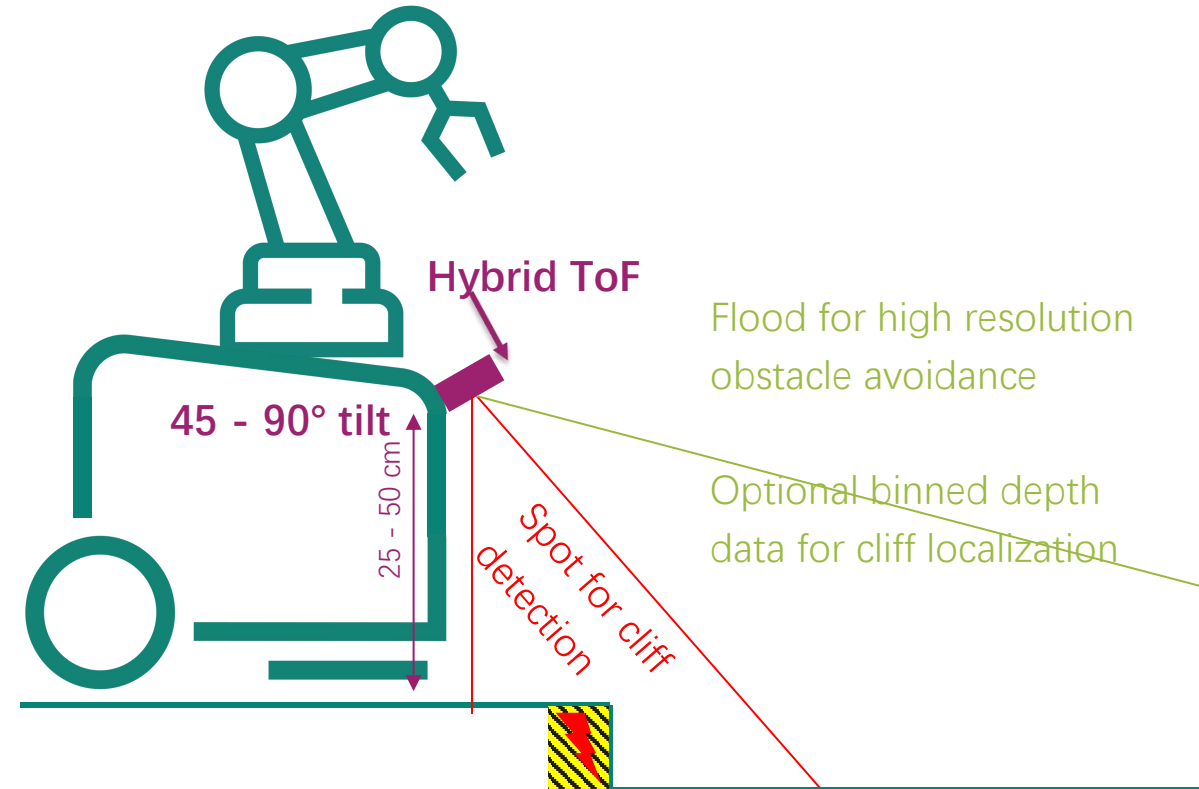
– Robot vacuum cleaners

- Key application: **SLAM & obstacle avoidance**
- Front: hybrid Time of Flight: **0° tilt**
- Optional side / rear: 2nd spot ToF to extend FoV: **0° tilt**



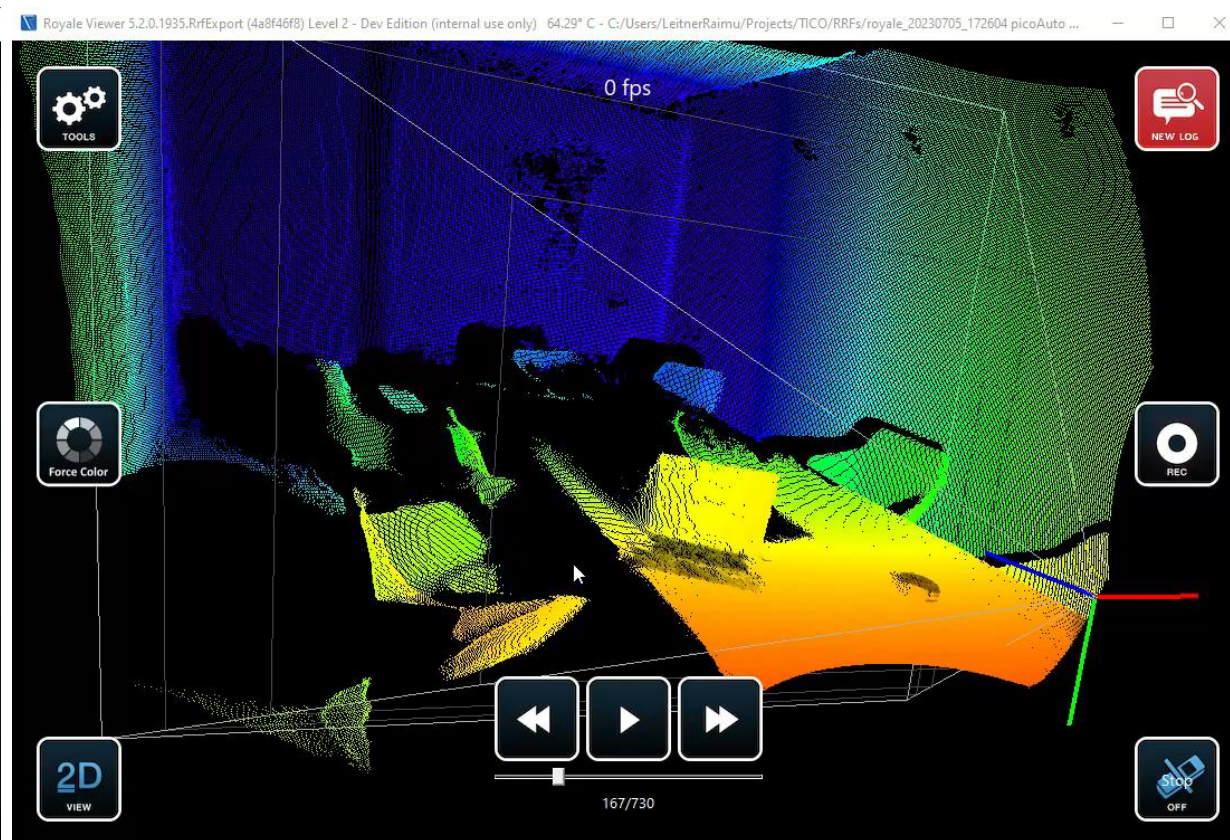
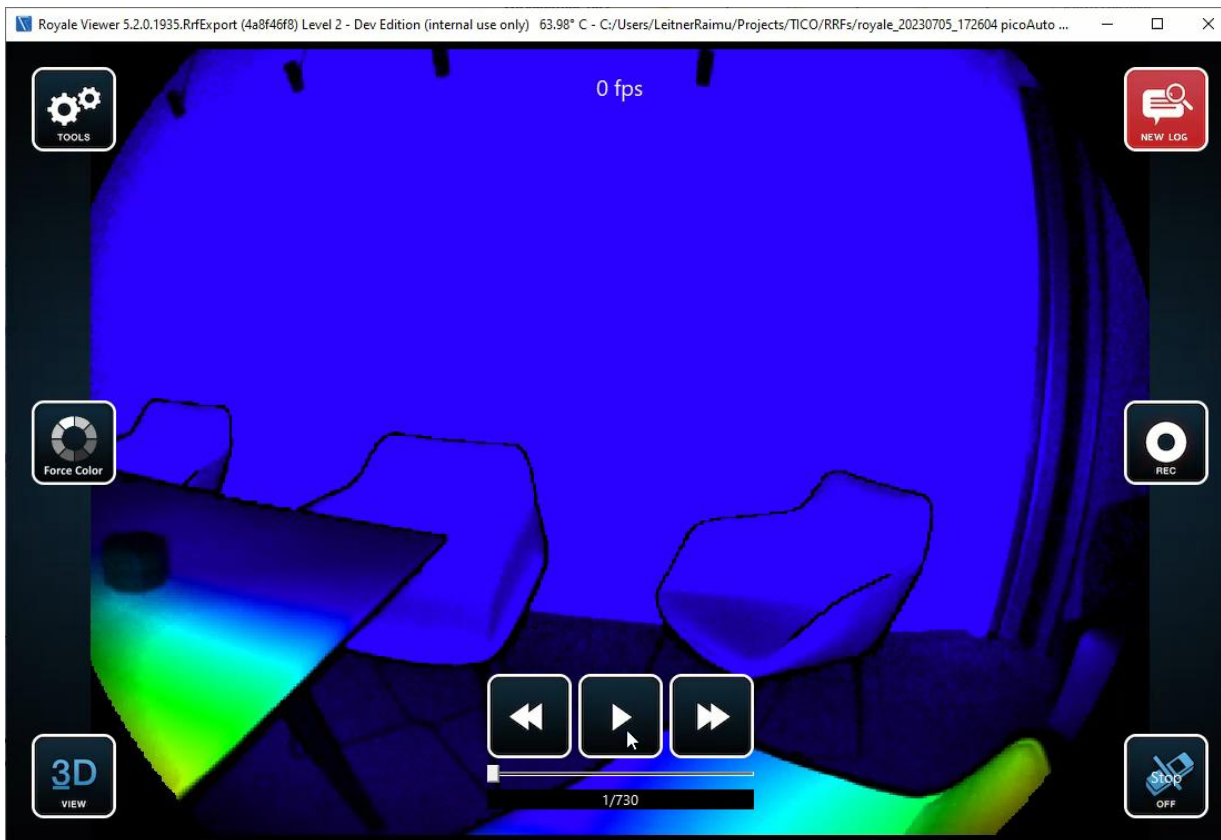
– Service Robots

- Key application: **obstacle avoidance & cliff detection**
- Front: Hybrid Time of Flight: **45 – 90° tilt**
- Flood and spot provide **redundant data for safety**



Pmd/Infineon IRS2877A - OMS picoAuto demo video

基于IRS2877A的汽车级的Demo视频



PicoAuto enables VGA resolution **long range** 3D mapping for most **accurate obstacle avoidance**

IRS2875C – hybrid Time of Flight Camera

One camera for two applications (SLAM + Obstacle avoidance)

hToF一个摄像头模组实现SLAM和避障（二合一的解决方案）

pmd

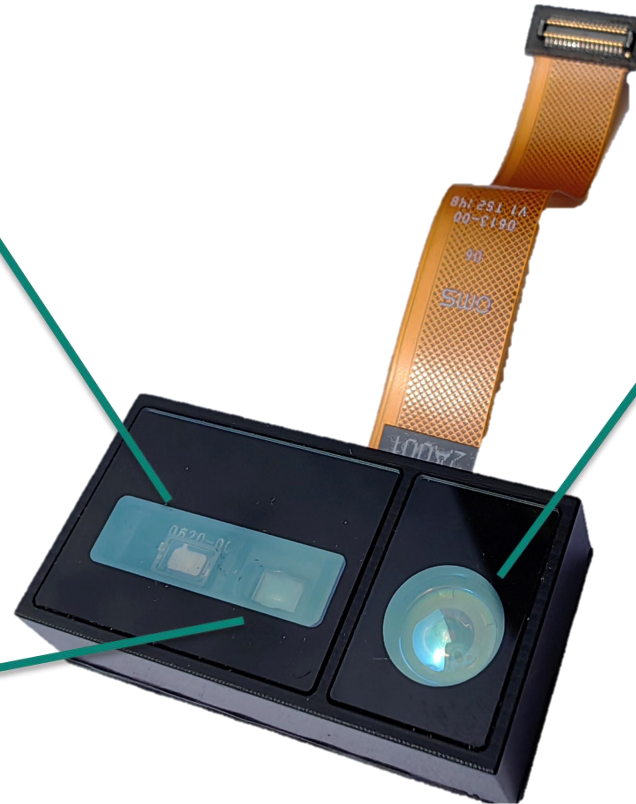


Flood illumination

Generating a high resolution
3D depth map
Ideal for advanced obstacle
avoidance

Spot illumination

Precise spot projection
generates high accuracy
long range depth points for
SLAM and 3D mapping



IRS2875C Time of Flight imager

High HQVGA (240 x 180 pixel)
resolution 3D depth imager
Highest integration at smallest
imager footprint allow for low 3D
camera BOM

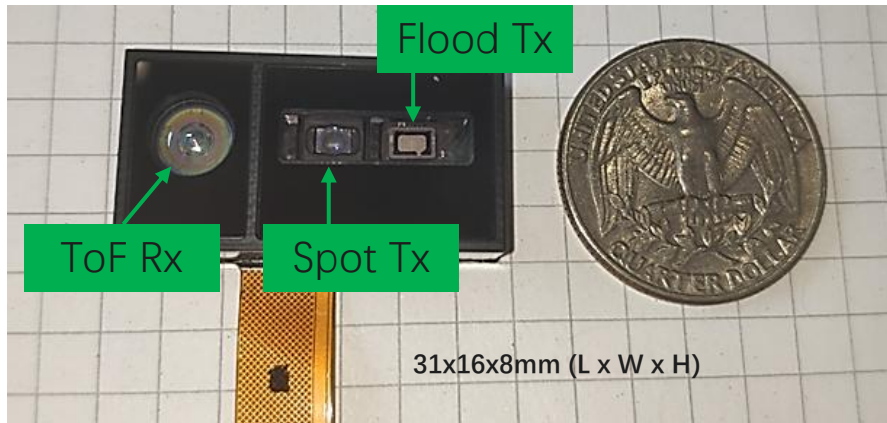
Dimension: 31 x 16 x 8 mm

Image: OMS DualTX / hToF OZT613 universal robotic camera
Using Infineon IRS2875C 3D ToF imager



hToF Module for Obstacle Avoidance AND Mapping

hToF 模组实现避障和建图



hToF module

=



Obstacle Avoidance (using Flood Tx)

+



Mapping (using Spot Tx)

– Idea

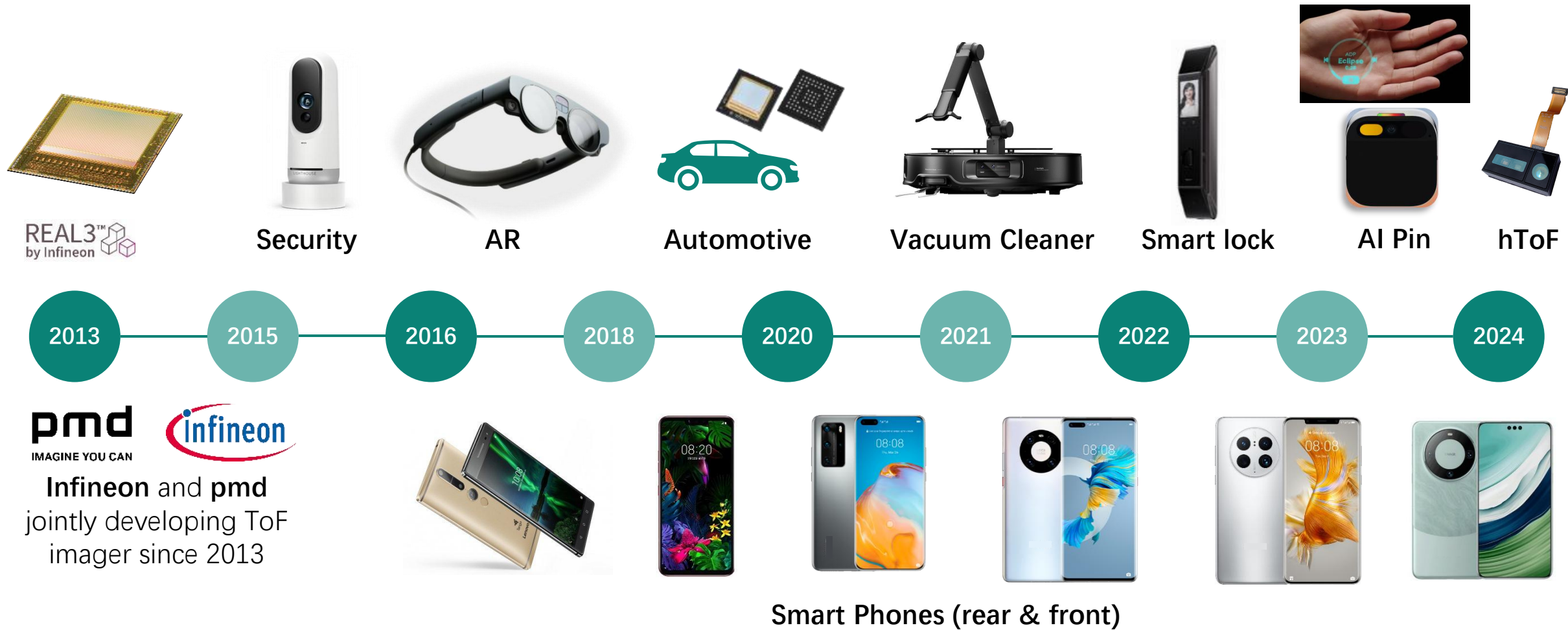
- Replace LDS for SLAM with competitive cost and slimmer design
- Use ToF **spot** system to match range and accuracy of LDS for **SLAM**
- Use ToF **flood** spot system for **obstacle avoidance**
- Do this with **one and the same module**

– Benefit

- SLAM + Obstacle avoidance in one solution
- Saving one sensor (LDS) in the device → **Reduced power, space, cost**

FOI = Field of Illumination

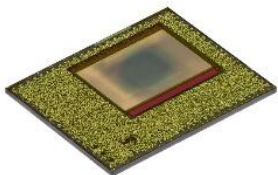
Infineon have a long history of successful 英飞凌在ToF市场深耕多年，是ToF市场领导者



Infineon / pmd **selling** into **millions** of **faceID systems** and **robots** each year!

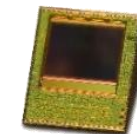
REAL3™ is the key for complete 3D smartphone solution

REAL3™ 提供给智能手机提供完整的3D解决方案



Rear facing ToF

- **Applications:**
 - enhanced video & photo **bokeh** / **auto focus**, fast **3D scanning**, immersive **AR**
- **3D camera illumination**
 - **spot** for lowest power and precise spot measurements
 - **flood** for dens measured depth map, e.g. fast scanning.
 - **hybrid** getting dot and flood in one
- **REAL3™ products:**
 - IRS2875C / IRS2975C HQVGA
 - VCSEL driver IRS9102C



Front facing ToF

- **Applications:**
 - Secure Face unlock and secure face payment, enhanced video & photo **bokeh** and rich **auto focus**, fast **3D scanning**, morphing/**AR**
- **3D camera implementation**
 - **under display** proven by Android **Biometrical Class3** (Strong) **certification**
- **REAL3™ products:**
 - IRS2877C / IRS2976C VGA
 - VCSEL driver IRS9102C
 - Android Level

Smallest cameras | Lowest power consumption
| Long range | Lower BOM cost | dot & flood

High VGA resolution | Smallest VGA imager | Highest
integration | Low module BOM | under display

3D Time-of-Flight is enabling the Smart Car – 3D ToF 传感器赋能智能汽车市场

In-cabin Sensing



ISO 26262
compliant

Driver Monitoring (DMS)

+ secure face-ID

- › Standard DMS + new services:
- › Payment (e.g. battery charging)
- › Seamless access to private data & services (e.g. music streaming)
- › Personalized car settings
- › HUD & mirror adjustment

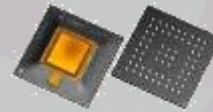


Occupant Monitoring (OMS)

- › Occupant detection & classification
- › Gesture control & body tracking
- › Smart airbags & seat belt detection



REAL3™
ToF imager
IRS2877A(S)



+

IRS9103A*
VCSEL driver

Short Range Exterior

Face-ID Car-Access

- › Hands-free Car Access (in combination with PKE avoiding relay attacks)
- › Activate personalized settings before and while entering the car

spoofing robust



Obstacle Avoidance

- › Automated doors
- › Near Field Lidar to cover blind spots in the ADAS sensor cocoon (cold start)
- › Autonomous parking



* In mass production, Jul 2024

